

Automatic Vehicle Number Plate Recognition

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Abstract

Automatic Vehicle Number Plate Recognition (ANPR) is a key technology in intelligent transportation systems (ITS), providing a foundation for automated vehicle monitoring, access control, toll collection, and law enforcement. As urban mobility and traffic density increase, the ability to automatically detect, recognize, and verify vehicle license plates in real time has become increasingly valuable. However, achieving robust ANPR performance remains challenging due to varying illumination, plate orientations, background clutter, and font styles. The goal of this project is to develop an efficient end-to-end ANPR system that can process both images and videos to detect, recognize, and verify vehicle number plates accurately.

1. Introduction

The proposed system is designed to automatically detect and recognize vehicle license plates from images or video frames. The system is divided into three main stages, each one ensures an accurate and reliable recognition.

a. License Plate Detection

In this stage the system identifies and localises the region of interest containing the license plate in the image or video frame given as input (Image 1). The plate region is isolated from the rest of the vehicle or background. A YOLO-based model is trained to achieve robust detection

performance, even under diverse environmental and lighting conditions.

[Figure removed - third-party image]

Image 1: License Plate Detection

b. Optical Character Recognition (OCR)

Once the plate has been detected and extracted, the alphanumeric characters are recognized using another model trained with YOLO. Instead of using traditional OCR tools, the use of YOLO for character-level recognition provides higher robustness and real-time performance.



Image 2: Character Detection

c. Character Arrangement and Reconstruction

In the final stage, a post-processing step arranges the recognized characters in left-to-right order according to their bounding box coordinates. This ensures that the final license number is reconstructed in the correct sequence, even if the characters are detected out of order.

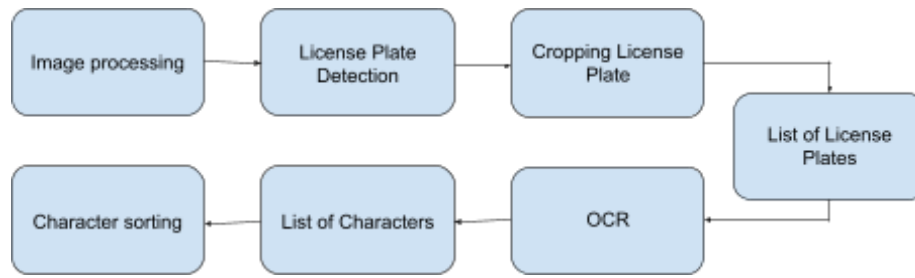


Image 3: Flowchart

2. Related Work

Automatic Number Plate Recognition (ANPR) systems have evolved significantly over the past decades, transitioning from traditional image processing techniques to modern deep learning-based architectures. Early methods relied on edge detection, color thresholding, and morphological operations to locate and segment number plates, followed by template matching or projection-based techniques for character recognition. Although effective in controlled environments, these systems were highly sensitive to illumination, rotation, and noise.

The introduction of classical machine learning approaches, such as Histogram of Oriented Gradients (HOG) and Support Vector Machines (SVM), slightly improved robustness by learning discriminative features, but still required precise segmentation and manual tuning. With the emergence of deep convolutional neural networks (CNNs), object detection frameworks like Faster R-CNN, SSD, and YOLO achieved real-time detection with high accuracy by processing entire images in a single pass.

In parallel, Optical Character Recognition (OCR) evolved from segmentation-based methods such as Tesseract to end-to-end sequence models like CRNN and Transformer-based architectures, capable of recognizing text without explicit character separation, albeit with high data and training complexity. To achieve a balance between

interpretability, efficiency, and accuracy, recent studies have employed YOLO-based OCR approaches that treat each character as an object detection problem. This design enables robust character recognition with lower data requirements and simpler implementation, forming the basis for the YOLO-based ANPR system developed in this project.

2.1 Integration and Real-World Application

Unlike most prior ANPR studies, which primarily focus on algorithmic performance, the system developed in this project bridges the gap between research and deployment

through an integrated, user-oriented design. The detection and recognition models are combined into a modular pipeline that processes both images and videos in real time.

To enhance accessibility, a Graphical User Interface (GUI) was implemented, allowing users to upload visual inputs, process them in real time, and verify recognition results. The GUI also provides a verification module, where users can input a specific license plate number and automatically receive feedback on whether the detected text matches, or differs from the reference value.

Overall, this project advances the field by transitioning from theoretical experimentation to real-world implementation, showcasing how a YOLO-based framework can deliver both high performance and operational usability for

intelligent transportation, parking management, and access control systems.

3. Dataset

The performance of ANPR systems largely depends on dataset diversity and annotation quality. In this project, two distinct datasets were employed.

The first dataset, obtained from Kaggle, consists of vehicle images containing visible license plates, and it was used to train the YOLOv8 number plate detection model. The second dataset, collected from Roboflow, includes labeled images of alphanumeric characters, employed for training the character recognition (OCR) model.

Both datasets underwent a preprocessing phase that includes resizing, normalization, and data augmentation to enhance the models' robustness to variations in illumination, orientation, scale, and background noise. This approach ensured that the trained models could generalize efficiently, achieving high accuracy in both plate detection and character recognition tasks.

3.1 License Plate Detection Dataset

For the plate detection stage, the dataset used was obtained from Kaggle and it contains approximately 2,000 vehicle images captured in daylight under various environmental conditions and angles. The diversity of the images ensures that the detection model can generalize efficiently to different plate sizes, positions, and perspectives. The dataset was annotated in YOLO format, with each image containing bounding box coordinates and a class label identifying the presence of a license plate.

3.2 OCR Dataset

The OCR stage relies on a dataset obtained from Roboflow, consisting of roughly 4000 cropped character images extracted from

vehicle plates. Each image corresponds to a single alphanumeric character, enabling the YOLO-based OCR model to detect and classify characters individually within cropped plate regions. Annotations are also provided in YOLO-compatible format, specifying both the class of the character and its bounding box.

3.3 Preprocessing and Data Preparation

Both datasets were preprocessed to ensure compatibility with the YOLO architecture. The main steps included:

- Converting all annotations to YOLO text format (class ID and normalized bounding box coordinates).
- Resizing all images to a standard resolution suitable for the models.
- Splitting the datasets into training (80%), validation (10%), and testing (10%) subsets to evaluate model performance accurately.

To enhance generalization, data augmentation techniques such as horizontal flipping, rotation, and brightness adjustments were applied.

4. Methodology

The proposed Automatic Number Plate Recognition system employs a three-stage approach, combining YOLO-based deep learning models with a user-friendly interface for practical deployment.

4.1 Three - Stage Pipeline

The system is structured into three sequential stages:

- Stage 1 – License Plate Detection:

In the first stage, a YOLOv8n model is trained to detect and localize vehicle number plates in images or video frames. The model receives the full vehicle image as input and outputs bounding boxes around detected plates. Its lightweight architecture

ensures high-speed inference suitable for real-time applications.

- **Stage 2 – Character Recognition (OCR):**
Cropped plate regions are processed by a YOLOv8l model trained to identify and classify individual alphanumeric characters. This approach replaces traditional OCR, offering robustness against variations in plate design and lighting.
- **Stage 3 – Character Arrangement:**
The OCR stage outputs multiple bounding boxes with class IDs for detected characters. To reconstruct the correct license number, a post-processing step arranges the characters from top to bottom and left to right, accommodating multi-line plates.

This approach ensures that the license number is reconstructed accurately, even when characters are slightly misaligned or multi-line plates are present.

4.2 Training Details

Both YOLO models were trained on the respective datasets described in the Data section. The main training configurations used in this work are:

- **Batch Size:** 16
- **Epochs:** 30 for plate detection, 50 for OCR
- **Annotation Format:** YOLO text files containing class ID and normalized bounding box coordinates
- **Data Splits:** Approximately 80% training, 10% validation, 10% testing

No significant data augmentation was applied beyond the standard YOLO preprocessing pipeline.

4.3 Graphical User Interface (GUI)

To make the system practical and user-friendly, a Gradio-based GUI was developed. The interface allows users to:

- Upload either an image or a video of a vehicle
- Click the Process button to run detection and OCR
- Reset the interface with the Reset button
- Optionally enter a vehicle number to verify if the detected plate matches

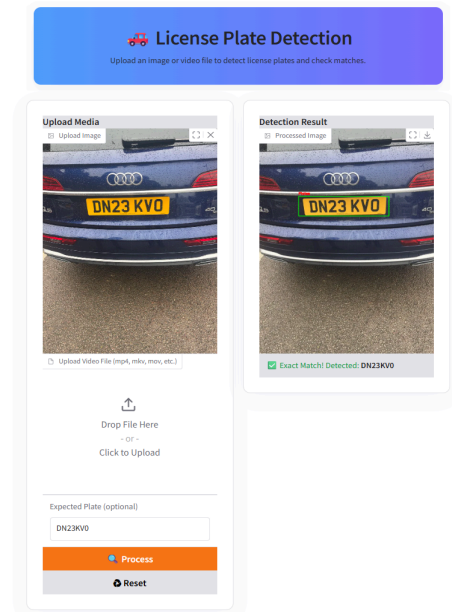


Image 4: GUI example

The output displays the vehicle image with bounding boxes around the detected plate and characters, along with a message indicating whether the entered number is completely matched, or unmatched.

The three-stage YOLO-based methodology provides a modular, interpretable, and efficient ANPR system. The separation of detection and OCR stages allows independent optimization and easier debugging. Post-processing ensures correct character sequencing, while the Gradio GUI demonstrates real-time usability and accessibility for end users. Together, these components deliver a robust and deployable system for automatic vehicle number plate recognition.

5. Experimental Results

The performance of the proposed YOLO-based Automatic Number Plate Recognition (ANPR) system was evaluated separately for the number plate detection and character recognition (OCR) stages. The experiments focused on measuring detection accuracy, recognition precision, and system robustness under various real-world conditions.

5.1 Evaluation Metrics

To assess the performance of both models, standard YOLO evaluation metrics were used, including Precision (P), Recall (R), and mean Average Precision (mAP) at different Intersection over Union (IoU) thresholds.

5.2 Number Plate Detection Results

The YOLOv8n model, used for number plate detection, demonstrated excellent performance with the following metrics:

- Precision (P): 0.998
- Recall (R): 1.0
- mAP@50: 0.995
- mAP@50–95: 0.975

These results confirm that the model reliably detects all visible number plates with minimal false positives. The high mAP values demonstrate strong localization accuracy across diverse lighting conditions, viewing angles, and backgrounds. The lightweight YOLOv8n architecture also ensures fast inference suitable for real-time applications.

5.3 OCR Results

For the character recognition stage the YOLOv8l model achieved the following results:

- Precision (P): 0.995
- Recall (R): 1.0
- mAP@50: 0.994
- mAP@50–95: 0.869

The OCR model accurately recognized the majority of alphanumeric characters, achieving high precision and recall. The slightly lower mAP@50–95 value reflects minor variations in bounding box alignment for individual characters, which is expected at small object scales. Nevertheless, the recognition performance remained highly consistent across different plate formats and environmental conditions.

5.4 Qualitative Evaluation

Visual inspections and testing on multiple images and videos confirmed that the system correctly identifies and reconstructs license numbers even when multiple vehicles appear in a single frame. Visual inspection confirmed accurate localization of both plates and characters, with post-processing successfully reconstructing the correct character order.

Only minor issues were observed in cases of severe motion blur or partial occlusion, where some characters were misclassified or missed. These limitations could be mitigated through dataset expansion and targeted augmentation techniques.

Overall, the experimental results validate the effectiveness of the proposed three-stage YOLO-based ANPR pipeline. The system demonstrates high precision, robustness, and interpretability, with efficient real-time performance suitable for intelligent transportation, traffic monitoring, and automated access control applications.

6. Discussion

The experimental analysis demonstrates that the proposed YOLO-based Automatic Number Plate Recognition (ANPR) system achieves near-perfect accuracy in both detection and character recognition stages. The YOLOv8n model for plate detection provides accurate localization with minimal false positives, while the YOLOv8l OCR model effectively

identifies alphanumeric characters even under varying lighting and viewing conditions.

The system's modular architecture, consisting of distinct detection and recognition stages followed by a post-processing step, allows for independent optimization and straightforward debugging. This separation not only simplifies training but also enhances interpretability, each stage can be visually verified by inspecting the corresponding bounding box outputs.

During testing on real-world images and videos, the pipeline maintained strong generalization and real-time performance. Most recognition errors occurred in challenging cases involving motion blur, partial occlusion, or low resolution, where one or more characters were occasionally misclassified. These limitations could be mitigated by expanding the datasets with motion-affected and low-light samples, or by incorporating temporal information from consecutive video frames.

Overall, the results confirm that the combination of lightweight YOLOv8n for detection and larger YOLOv8l for OCR offers a strong trade-off between inference speed and accuracy, making the proposed system suitable for real-time deployment in intelligent transportation and monitoring scenarios..

7. Conclusion

In this project, a complete end-to-end ANPR system was developed using YOLOv8 models for both plate detection and optical character recognition. The three-stage design ensures a modular, interpretable, and highly accurate pipeline.

Experimental results demonstrate that the system performs reliably across diverse lighting and environmental conditions, achieving near-perfect detection and recognition rates. Moreover, the integration of a graphical user interface (GUI) allows

seamless interaction, enabling users to upload images or videos, process them in real time, and verify recognition results.

By bridging the gap between research and application, the proposed framework demonstrates that YOLO-based architectures can effectively replace traditional OCR pipelines while maintaining real-time performance and ease of deployment.

7.1 Future Work

Future work will focus on extending the dataset to include plates from different countries and environmental conditions, optimizing inference speed for edge devices such as Raspberry Pi or NVIDIA Jetson Nano, and improving robustness under low-light or motion-blur scenarios. Additional efforts will also explore integrating the system with cloud-based APIs and smart mobility platforms, enabling large-scale, data-driven traffic management applications.

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